A Class-AB Audio Power Amplifier: Design, SPICE Verification and PCB Layout

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Report type	Technical report
Status	Schematic, SPICE simulation and PCB layout complete; DRC passed; Gerber and drill files generated. Fabrication, assembly and bench measurement are not reported
Tools	KiCad, LTSpice
Date	2026

Abstract. A discrete class-AB audio power amplifier was designed using the three-stage Lin topology: a long-tailed input pair with current-mirror load, a voltage-amplifier stage (VAS) with Miller compensation, and a complementary emitter-follower output stage biased by a thermally coupled V_{BE} multiplier. Global feedback sets a closed-loop gain of 23 (27 dB). The design targets ±25 V rails and an 8 Ω load. LTSpice operating-point, transient, AC and Fourier analyses were used to check biasing, crossover behaviour and frequency response before layout; the simulated response is flat at 27 dB from approximately 20 Hz to 50 kHz with a phase margin above 60°. The board was laid out in KiCad with star grounding and separated signal and power returns, passed design-rule checks, and was exported as Gerber and drill files. Performance has been established by calculation and simulation only.

Index terms: class-AB amplifier, Lin topology, negative feedback, Miller compensation, V_{BE} multiplier, crossover distortion, PCB layout.

1. Objectives and Requirements

Class-AB operation was chosen as a compromise between class-A linearity and class-B efficiency: biasing the output pair slightly into conduction removes the class-B dead zone at the cost of quiescent dissipation and a bias that must be stabilised against temperature.

Table 1. Requirements.

Requirement	Target	Basis
Supply	±25 V	Design constraint
Load	8 Ω	Design requirement
Closed-loop gain	27 dB (x23)	Design target
−3 dB bandwidth	20 Hz to 50 kHz	Design target
Phase margin	> 60°	Design target
Output power	Approx. 40 W (ideal rail limit 39 W)	Calculated upper bound

2. Topology

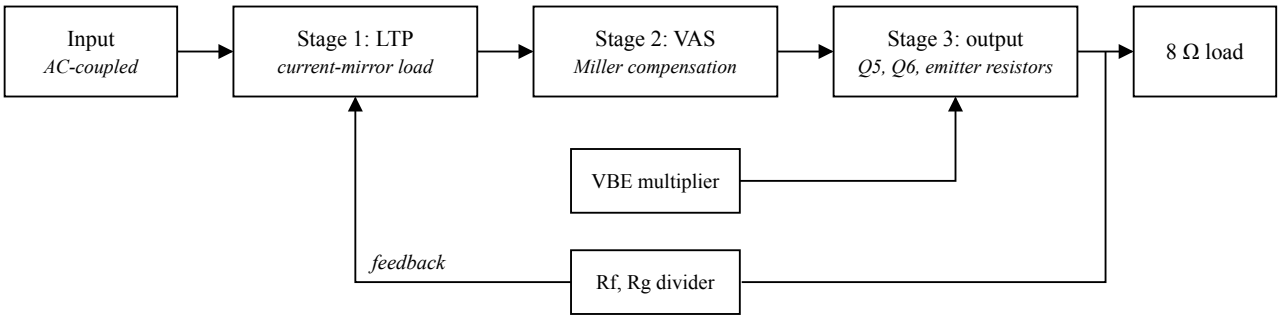


Figure 1. Signal path of the three-stage amplifier. The long-tailed pair (LTP) converts the difference between input and fed-back output into a current; the VAS provides most of the open-loop voltage gain; the complementary emitter followers provide current gain. R_f/R_g closes the global loop and sets the DC output at mid-rail.

3. Design Analysis

3.1 Gain and feedback

$$A_v = 1 + R_f/R_g, \quad \beta = R_g/(R_f + R_g), \quad T = A_{OL}\beta \tag{1}$$

With $R_f = 22\ k\Omega$ and $R_g = 1\ k\Omega$, $A_v = 23$ (27.2 dB). Distortion, output impedance and gain error are reduced by approximately (1 + T) [1], [3].

3.2 Input stage

$$g_{m1} = I_{tail} / (2V_T) \tag{2}$$

For $I_{tail} = 2$ mA and $V_T = 25.9$ mV, $g_{m1} \approx 38.6$ mS. The tail current sets input transconductance, loop gain and slew rate.

3.3 Bias, crossover and thermal stability

$$V_{bias} = V_{BE}(1 + R_2/R_1) \approx 2V_{BE} + 2I_qR_e \tag{3}$$

The V_{BE} multiplier holds the output pair at a quiescent current I_q. Since V_{BE} falls by about 2 mV/°C, a fixed bias would cause I_q to rise with temperature. Two measures counter this: emitter resistors R_e provide local feedback, and the bias transistor is mounted on the output heatsink so that V_{bias} tracks device temperature. Oliver's condition, $I_qR_e \approx V_T$, gives an optimum near 118 mA for $R_e = 0.22\ \Omega$ [3]. The design uses approximately 30 mA, accepting slightly higher crossover distortion for lower quiescent dissipation.

3.4 Compensation and slew rate

$$SR = I_{tail} / C_m, \quad f_{FP} = SR / (2\pi V_{pk}) \tag{4}$$

Miller capacitance C_m across the VAS sets the dominant pole so that unity loop gain occurs before the output-stage and load poles add significant phase lag. The same capacitor limits slew rate: with 2 mA and 100 pF, SR = 20 V/µs, giving a full-power bandwidth of about 127 kHz at 25 V peak, above the audio band.

3.5 Dissipation and safe operating area

$$P_{dev,max} \approx V_{CC}^2 / (\pi^2 R_L), \quad \theta_{sa} \leq (T_{j,max} - T_a)/P_{dev,max} - \theta_{jc} - \theta_{cs} \tag{5}$$

For a sinusoid into 8 Ω at ±25 V, worst-case dissipation per device is about 7.9 W, plus quiescent dissipation $V_{CC}I_q \approx 0.75$ W. These figures size the heatsink and copper area. Reactive loads increase instantaneous dissipation beyond the resistive case.

4. Design Point

Table 2. Design point. Component values follow common discrete class-AB reference designs and were checked against the SPICE operating point.

Block	Value or part	Evidence
Input pair	2N5551 / 2N5401	Design choice
Tail current	2 mA	Design choice
VAS	2N5551, C_m = 100 pF	Design choice
Bias	V_{BE} multiplier (2N5551) on output heatsink	Design choice
Output pair	TIP41C / TIP42C	Design choice
Emitter resistors	0.22 Ω, 5 W	Design choice
Quiescent current	Approx. 30 mA per device	Design choice
Feedback network	R_f = 22 kΩ, R_g = 1 kΩ	Design choice
Output network	Zobel (approx. 10 Ω, 100 nF); 1 µH inductor shunted by 10 Ω	Design choice
Input impedance	Approx. 22 kΩ	Calculated
Slew rate	20 V/µs	Calculated
Peak dissipation per device	7.9 W	Calculated

5. Simulation

The signal path was simulated in LTSpice with manufacturer transistor models: DC operating point (bias and mid-rail output), transient analysis of the crossover region, AC analysis of the closed-loop response, and Fourier analysis at 1 kHz.

Table 3. Analytical and simulated results.

Quantity	Analytical	Simulated	Measured
Midband gain	27.2 dB	27 dB	Not measured
−3 dB bandwidth	Target 20 Hz to 50 kHz	Approx. 20 Hz to 50 kHz	Not measured
Phase margin	Target > 60°	> 60°	Not measured
Crossover region	N/A	Continuous transfer characteristic when biased	Not measured

Numerical THD results from the Fourier analysis are not reported in this document.

6. PCB Layout

- **Grounding.** Small-signal ground (input stage, feedback divider) and power ground (output return, decoupling) are joined at a single star point near the supply.
- **Routing.** Supply, output-emitter and loudspeaker paths use short, wide traces or pours; small-signal traces are kept short and away from the output loop.

- **Decoupling.** Reservoir and high-frequency capacitors are placed at the output-stage collectors.
- **Thermal.** Copper pour under the output devices and bias transistor provides a return path and heat spreading.
- **Manufacturing outputs.** Footprints were checked against manufacturer land patterns; the board passed DRC and Gerber and Excellon drill files were exported [6].

7. Unevaluated Targets

Table 4. Design targets for future bench validation.

Metric	Target	Status
THD+N at 1 kHz, 1 W	< 0.05%	Not evaluated
THD+N at 20 kHz	< 0.2%	Not evaluated
Damping factor (8 Ω)	> 100	Not evaluated
PSRR at 100 Hz	> 60 dB	Not evaluated

8. Limitations

- No hardware was assembled or measured; all performance figures are analytical or simulated.
- Stability was assessed with a resistive load; behaviour with reactive loudspeaker loads was not simulated.
- Junction temperature and SOA were checked by calculation only.

9. Conclusion and Future Work

The analytical design and simulation agree on gain, bandwidth and phase margin, and the layout is ready for fabrication. Future work: assemble the board and measure gain, bandwidth, THD against power and frequency, and quiescent current drift; simulate reactive loads; and evaluate two-pole compensation to increase slew rate at a given phase margin.

Appendix A. LTSpice Netlist (excerpt)

```
Vcc vcc 0 25
Vee vee 0 -25
Vin in 0 AC 1 SIN(0 20m 1k)
Q1 n1 in te 0 2N5551        ; input long-tailed pair
Q2 n2 fb te 0 2N5551
Itail te vee 2m
Q3 vas n1 vee 0 2N5551      ; VAS
Ivas vcc vas 5m
Cm vas n1 100p              ; Miller compensation
Q4 vas nb vas 0 2N5551      ; VBE multiplier
Rb1 vas mid 4k7
Rb2 mid nb 2k2
Q5 vcc vas oe1 0 TIP41C     ; output pair
Q6 vee nb oe2 0 TIP42C
Re1 oe1 out 0.22
Re2 oe2 out 0.22
Rf out fb 22k               ; feedback network
Rg fb 0 1k
Cout out spk 1000u
Rload spk 0 8
.tran 5m
.ac dec 20 10 1meg
.four 1k V(spk)
```

Listing A1. Representative netlist excerpt (models from manufacturer .lib files).

References

[1] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 8th ed. Oxford University Press, 2020.

[2] P. Horowitz and W. Hill, *The Art of Electronics*, 3rd ed. Cambridge University Press, 2015.

[3] D. Self, *Audio Power Amplifier Design*, 6th ed. Focal Press, 2013.

[4] B. Cordell, *Designing Audio Power Amplifiers*, 2nd ed. Routledge, 2019.

[5] H. C. Lin, "Quasi-complementary transistor amplifier," *Electronics*, vol. 29, pp. 173-175, 1956.

[6] KiCad EDA, "PCB design and design rule checking," documentation, kicad.org.